Equity of Chicago Youth Programs Data Analysis

STAT 303-1 Data Science I with Python - Final Report

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2024-12-09

Our analysis aims to examine and determine the relationships between equity levels in My CHI. My Future. Programs and four key factors which are neighborhood and associated socioeconomic status, criminality, program subject, and age level/academic performance. After extensive data analysis, we make the following recommendations. Program costs in low-income areas should be decreased or eliminated entirely. Neighborhoods farther from downtown/central Chicago should see increased transportation offerings for their programs. Programs in high-crime neighborhoods should be strategically relocated during high-crime seasons to avoid conflict and violence. My CHI should create programs tailored to specific community's preferences in terms of program types. Additionally, Chicago needs more academic programs, specifically math programs in order to see increased test scores in STEM-related subjects. The number of academic programs for children in Southside communities should be increased, and the quality of academic programs for teenagers all across the city should be increased.

1 Problem statement

We seek to examine and determine the relationships between equity levels in Chicago Youth Programs and four key factors: neighborhood and associated socioeconomic status, criminality, program subject, and age level/academic performance. Specifically, the four questions we will explore are:

1. How do trends of price, transportation, and free food appear between neighborhoods? Are these trends equitable?
2. Are there any correlations between crime and availability of programs in certain districts?
3. Are there uneven distributions of subject matter in the programming?
4. Are programs equally available across different age groups? How do academically-focused programs affect academic performance?

2 Data sources

1. My CHI. My Future. Programs

The My CHI. My Future. Programs dataset contains information on youth programs across the city of Chicago, including a broad range of program types. The dataset contains information on over 100,000 programs, and includes a variety of program types such as academic, sports, music, career, and many others. This will be the main dataset that we use in this report, and it will be used in every EDA. This data contains many missing values, which will be dealt with in the following sections.

2. Chicago Boundaries Data

This dataset contains geographic information for the boundaries of the 77 Chicago communities. It will be used to plot the different Chicago neighborhoods on geospatial plots.

3. Chicago Socio-Economic Indicators by Neighborhood from 2010 Census

This dataset contains six socioeconomic 'indicators' such as per capita income and employment information for each of the Chicago community areas.

4. Chicago Crime 2022 Dataset

This dataset contains information on specific crimes that have happened in Chicago. Information for each crime includes the date/time, location, and type of crime.

5. Chicago Public Schools 2018-19 Progress Report Data

This dataset contains report cards for each elementary schools, middle schools, and high schools in the Chicago area from 2019 (this is the most recent data available).

3 Data quality check / cleaning / preparation

3.1 My CHI. My Future. Dataset

Numeric Variables Distributions

	Capacity	Min Age	Max Age	ZIP Code	Latitude	Longitude
count	2.207110e+05	227746.000000	227746.000000	220353.000000	219615.000000	219615.000000
mean	5.655422e+03	8.686647	44.162668	60629.190104	41.851651	-87.680025
std	7.016113e+05	6.518894	42.095325	27.371589	0.099947	0.118872
min	0.000000e+00	0.000000	0.000000	60018.000000	38.922466	-120.961998
25%	1.000000e+01	3.000000	12.000000	60617.000000	41.776459	-87.717003
50%	1.500000e+01	6.000000	18.000000	60628.000000	41.863098	-87.680382
75%	2.800000e+01	13.000000	99.000000	60641.000000	41.945400	-87.638603
max	9.910181e+07	65.000000	171.000000	66210.000000	42.147499	-87.530502

Some of the minimum and maximum values, such as for variables `Max Age`, `Min Age`, `Latitude`, `Longitude`, and `Capacity` seem unreasonable, and will be dealt with in upcoming sections.

Variables Count, Unique Values, and Data Type

Column Name	Count (Non-Null)	Number of Unique Values	Column Data Type
Program Name	227746	30491	object
Description	227746	42775	object
Category Name	227745	23	object
Capacity	220711	195	float64
Min Age	227746	34	int64
Max Age	227746	63	int64
ZIP Code	220353	82	float64
Program Type	227746	1	object
Start Date	227746	1333	object
End Date	227746	1302	object
Start Time	203319	161	object
End Time	203302	276	object
Geographic Cluster Name	216610	89	object
Program Price	227746	4	object
Scholarship Available	227746	2	bool
Latitude	219615	1659	float64
Longitude	219615	1562	float64
Transport Provided	224493	2	object
Has Free Food	225996	2	object
Meeting Type	227746	2	object
Tag	227746	4	object

There are many variables with missing values (such as `Longitude`, `Latitude`, and `Geophraphic Cluster Name`), which will be either imputed or removed depending on the context, in the data cleaning process.

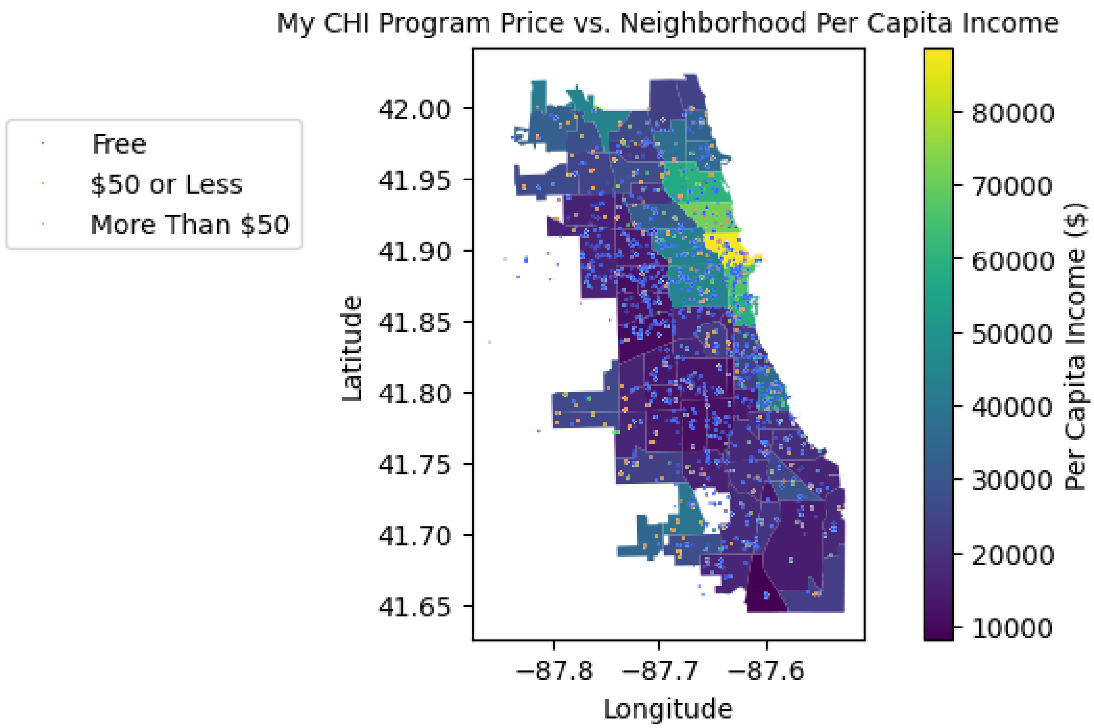
Distributions of other datasets used in this analysis can be seen in the appendix. Since each EDA uses the data differently, the issues that we ran into with all data will be discussed and dealt with in the individual EDAs.

4 Exploratory Data Analysis

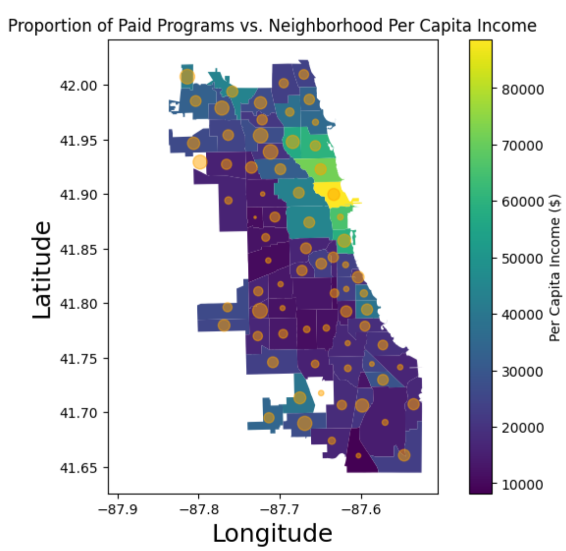
4.1 Analysis 1: How do trends of price, transportation, and free food appear between neighborhoods? Are these trends equitable?

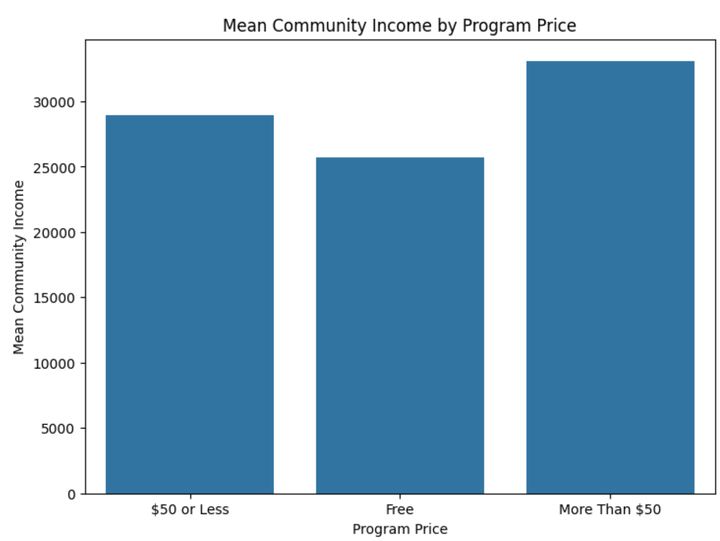
By Nolan White

A natural first step to visualize this EDA might be to make an income heatmap of the areas, and then overlay pricing information for every organization. This produces the geospatial map found below. The issue with this approach is that on the surface, there does not seem to be much of a trend between program price and the area in which it is offered. The dots, one for every individual organization, do not seem to form any clear pattern. With more than 200,000 organizations in the My CHI dataset, there is no easy way to get a sense of the trends across the entire dataset plotting every organization.



To get around this issue, we decided to compare the proportion of paid organizations across neighborhoods, rather than display each individual organization's price. This simplifies the visualization while making trends clearer. The resulting graph and barplot showing the mean community income across the three price categories are below.



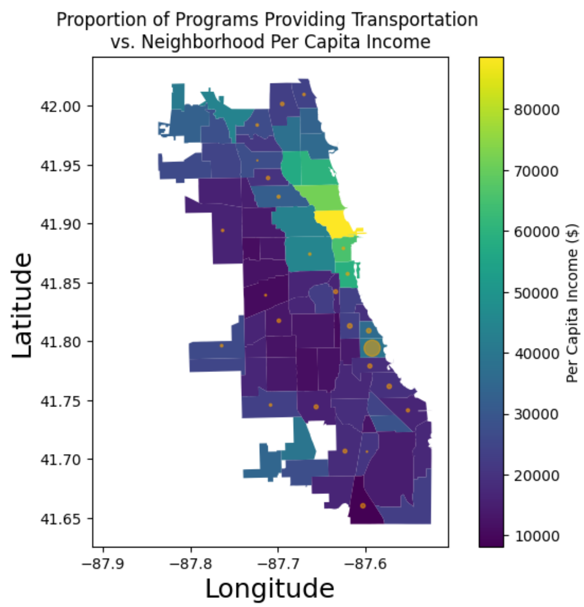


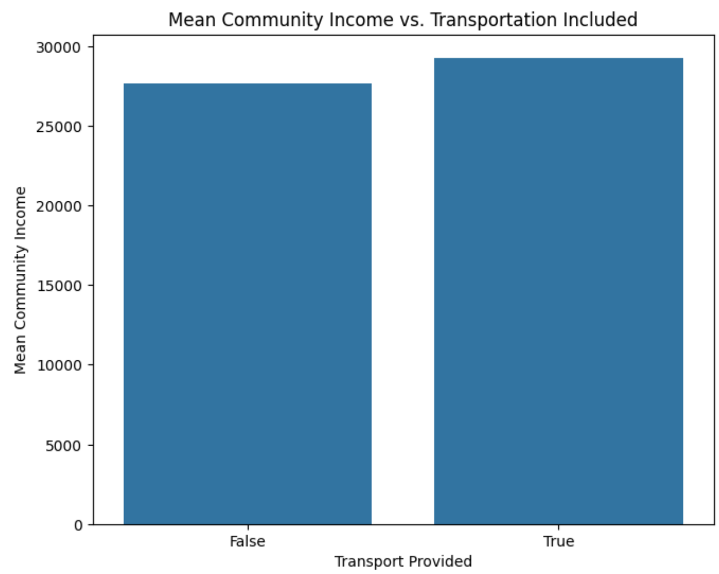
```
Correlation Coefficient 0.348
```

This barplot shows that the overall trend is that more expensive programs are located in areas with higher per capita incomes, which is the trend we would want to see in an ideal world. Although this general trend indicates that program pricing is headed in the right direction, the correlation between price and income is only 0.348, meaning the relationship is not as strong as it could be.

Furthermore, the trend does not hold up when inspecting outliers. Take for example West Elsdon, where ~93% of programs offered are paid, the highest of proportion across all neighborhoods, and the per capita income in this area is only $15,754, significantly below the city average. Many other low-income neighborhoods also have a disproportionately large number of paid programs.

While the trend of program price increasing with income across neighborhoods is equitable and what we would hope to see, inspection of the outliers shows there is significant room for improvement.



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Correlation Coefficient 0.083
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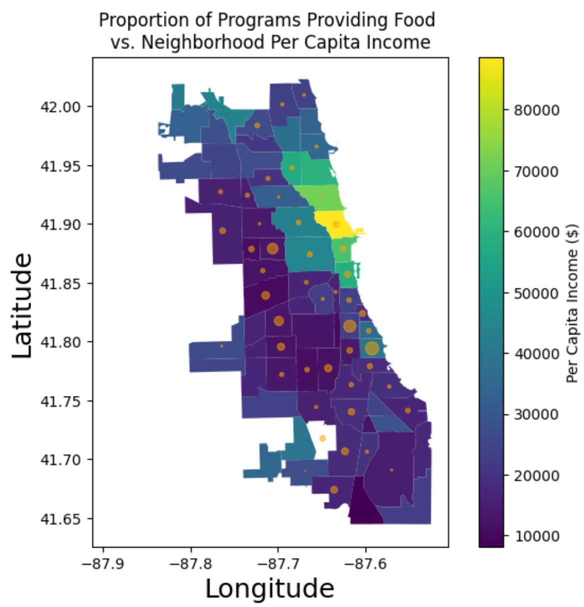
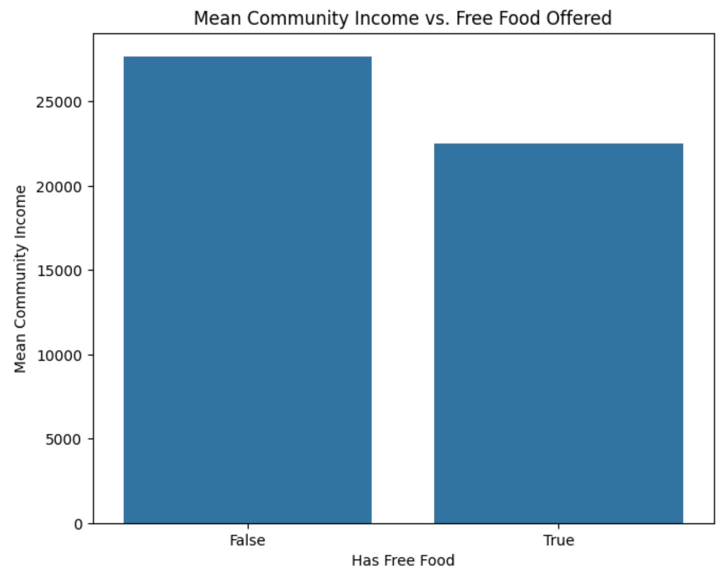
Our analysis of transportation offerings uses similar visualizations as for program price.

It can be concluded from the barplot that the mean community income for organizations providing transportation is slightly higher than the income for organizations that do not provide transportation. However, the correlation between neighborhood income and transportaiton provided is quite low at 0.083, meaning the linear relationship between these variables is pretty weak. Additionally, the geospatial plot does not seem to provide strong evidence for this relationship or reveal any different trends.

A notable outlier on the map is the region around the South Lakefront and Hyde Park neighborhoods, which have the largest proportion of programs offering transportation by a large amount. One explanation for this might be that these organizaitons are tied to the nearby Museum of Science and Industry or the University of Chicago. The most important takeaway from analyzing the availablity of transportation among My CHI programs is the widespread lack of transportation offerings. Out of 224,493 organizations reporting information on transportation amenities, only 157 programs actually provide transportation for their participants.






```
Correlation Coefficient -0.013
```

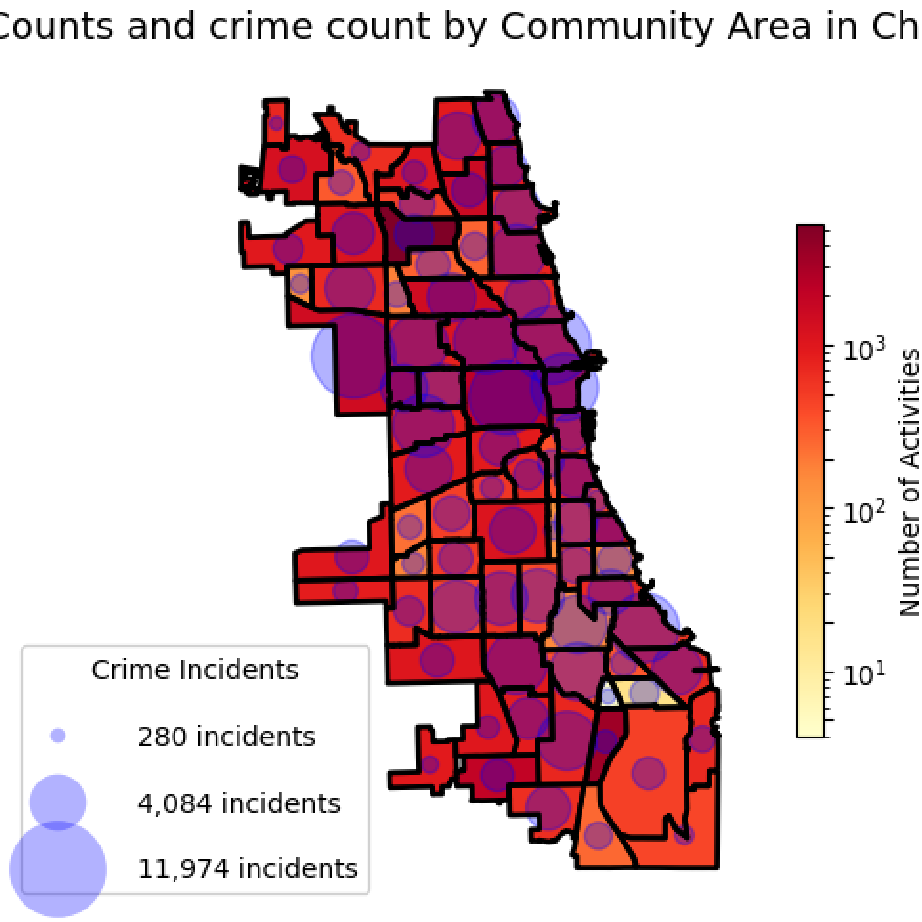
The barplot appears to show quite a significant difference, ~$5000, between community income for programs providing free food and programs that do not provide free food. This would suggest that programs in lower income areas have a higher proportion of free food. The geospatial plot supports this relationship, as some of the darkest communities have some of the largest proportions of programs offering food. However, because the correlation between neighborhood income and the proportion of free food provided is only -0.013, this relationship is not notably strong.

To sum up the key findings after exploring these three variables, there seems to be evidence that the price of programs and the availability of food are trending in an equitable direction. Cheaper programs are more commonly found in areas with lower community income, and free food is more often provided in areas with lower community income. Although quite rarely offered, programs providing transportation on average have a higher community income than those which do not provide transportation. Even though the transportation variable does not seem to correlate strongly with income levels, there is definitely room to improve equity with transportation amenities by making transportation more widely available in lower-income areas.

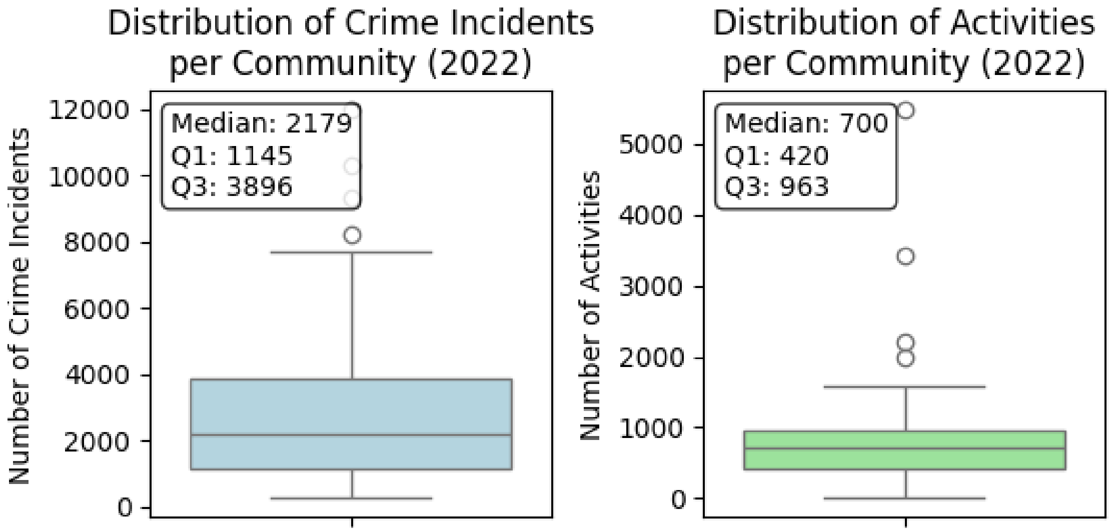
4.2 Analysis 2: Are there any correlations between crime and availability of programs in certain districts?

By Pedro Nieto Rocha

To examine the impact of crime on the Chicago Communities, our initial first step was to exploratorily visualize the number of programs in each of the Chicago communities, and compare to the number of crime incidents for the year 2022. By using geopandas to impute the locations of the programs within the Chicago communities, we created the following visualizations that shows this data side by side.

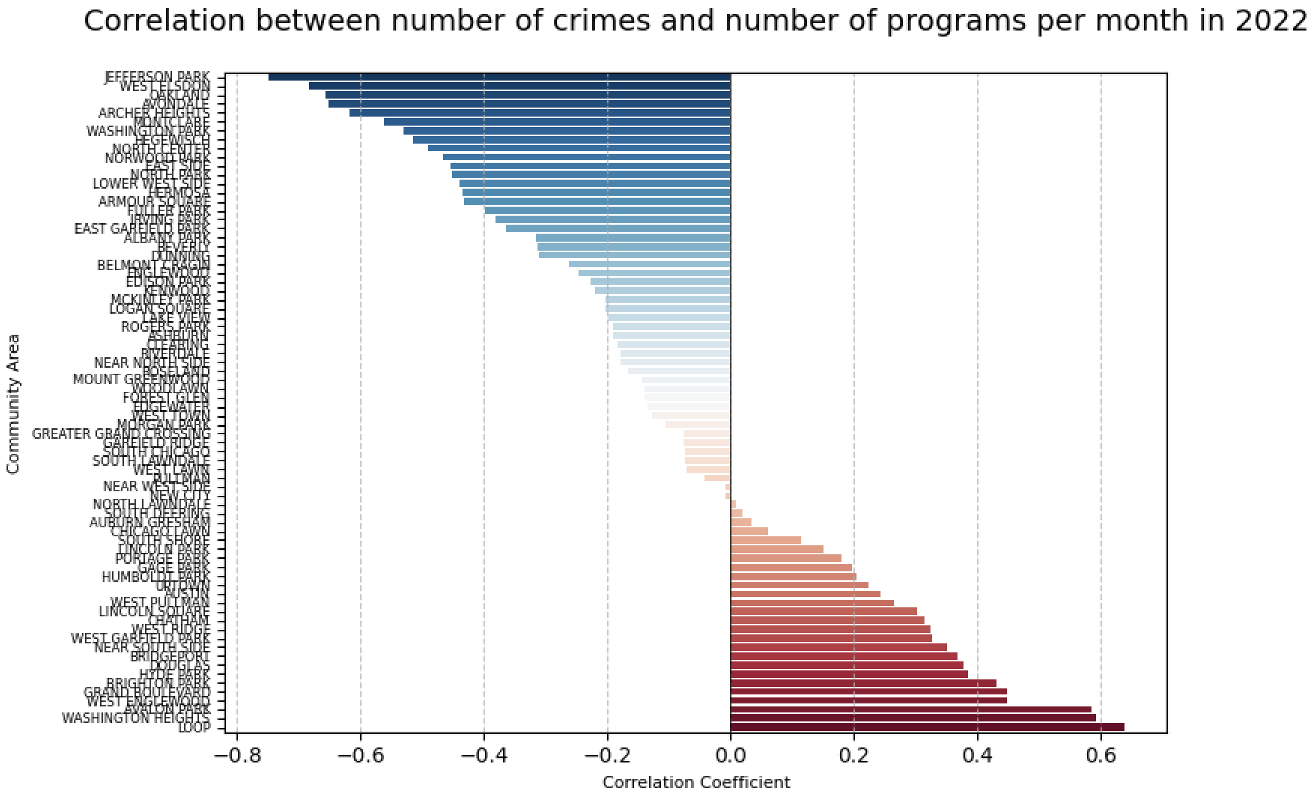


In the visualization we can see that there seems to be no apparent correlation between the number of activities and crime incidents in 2022, but we can see that the higher magnitudes of crime are mostly in the southern and northwest communities. What did catch our eye was that for both the number of crime incidents and the number of activities there is a difference of about two magnitude between some of the most violent and most prosperous communities, and for the communities with the highest number of activities.



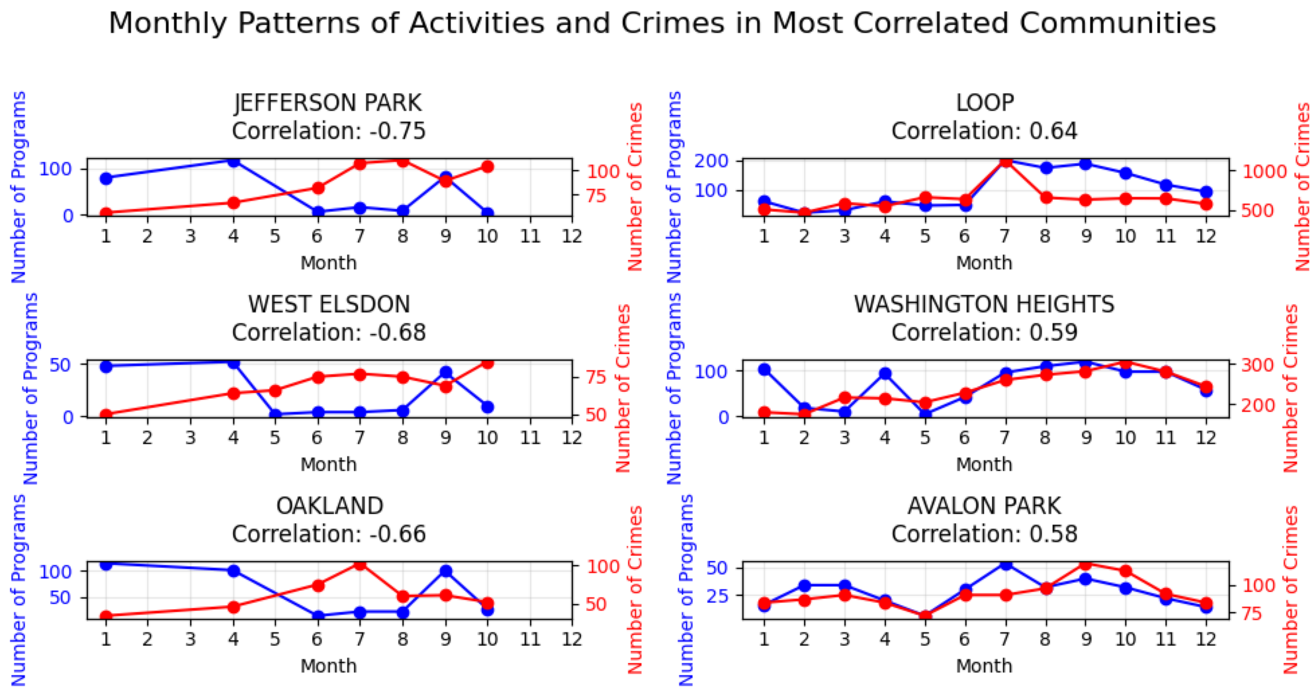
To further visualize the disparity in safety and program access, this boxplot helps to see just how many orders of magnitude the outliers are in terms of crime and number of activities. Among the outliers for crime (with respective crime count), communities such as Austin (11974), Near North Side (10335), Near West Side (9304) and South Shore (8213) are about two to three times more violent than the average of crime in communities (3051 incidents, and median 2179). With respect to the activity data, places such as Irving Park (5482), Pullman (3417), Near West Side (2206) and Morgan Park (1974) are up to 6 times the mean activity count (812.5 activities and median 700). Data for these outliers and averages is in the appendix.

In an effort to further investigate the relationship between crime and the number of activities, we grouped the number of incidents per month in 2022, as well as the number of activities per month and computed a total correlation coefficient, which turned out to be -0.088, but upon drilling down on the correlations between crime and number of activities per district we discovered that in some communities crime and programs do have a meaningful relationship.



In this plot, we can see that broken down into monthly frequency, there are many strong negative correlations between crime and number of activities as well as some moderately positive and negative correlations. By dropping calumet heights which had a misleading negative correlation of -1 due to missing data points on most of the year, we view quite a compelling picture for some crime ridden communities in terms of lack of access to programs.

To further analyze the relationship between crime and number of programs, we can drill down into the individual plots that peer into the possible seasonality and relationship of crime per month for the top 3 strongest positive and negative correlations.



As we can see, there are clear trends in the seasonality of crime in relation to the number of programs available. The most negatively correlated communities (Jefferson Park, West Elsdon, and Oakland) show an inverse relationship where program activity tends to decrease as crime increases, with programs typically peaking in spring months while crime rises during summer and fall. This pattern is particularly pronounced in Jefferson Park, where program activity drops sharply as crime rates climb. In contrast, positively correlated communities like the Loop, Washington Heights, and Avalon Park demonstrate synchronized movements between programs and crime rates, with both metrics often peaking during later months and showing similar seasonal patterns. The Loop, notably, has the highest overall numbers of both programs and crimes, with concurrent peaks around August. While these correlations are clear, they don't necessarily indicate causation; the relationships could be influenced by various factors such as weather patterns or population density.

For the previous reasons, it is productive to conduct statistical hypothesis testing to determine which of the outliers are sufficiently correlated to determine that it might be more reasonable to suspect causation. That is, of statistical significance $\alpha = 0.05$

For testing the relationship between crime and program availability:

$$H_0 : \rho = \mu$$

$$H_1 : \rho \neq \mu$$

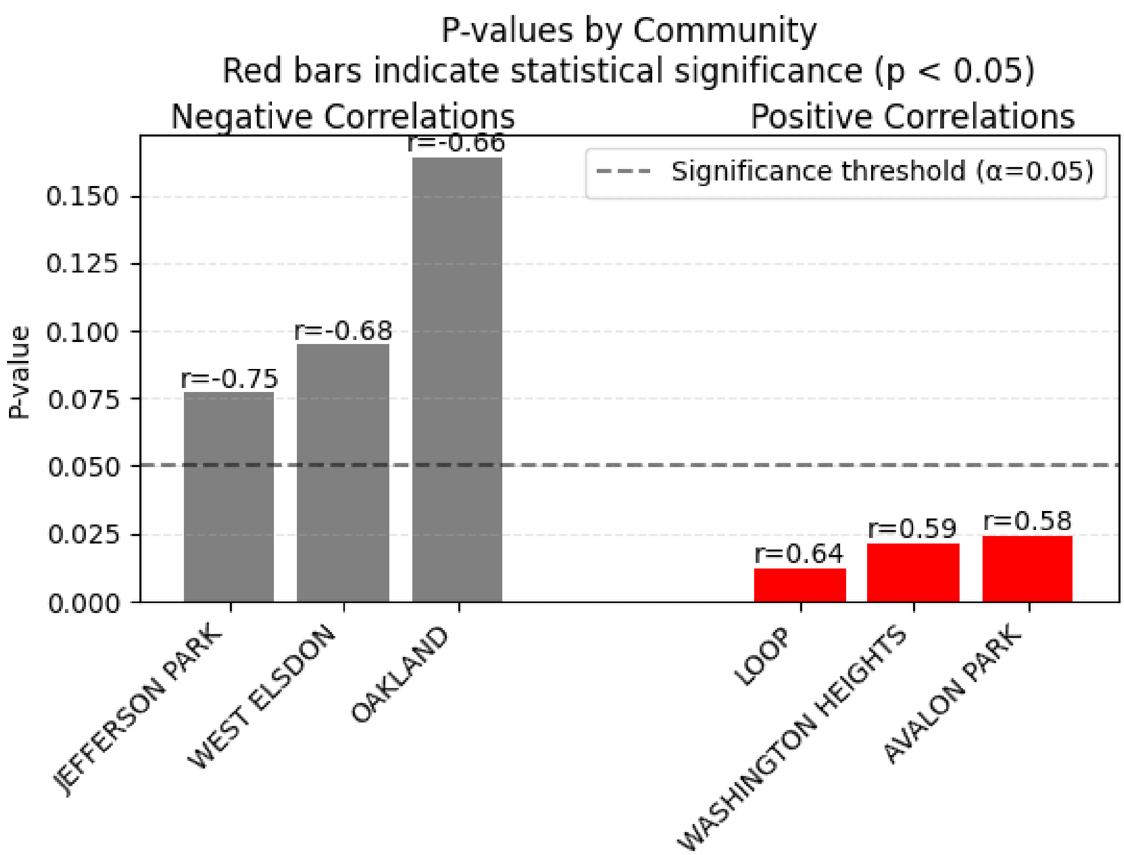
Where ρ represents the correlation coefficient for an individual community and μ represents the global mean correlation (-0.088)

The statistical analysis examines whether the observed correlations in selected Chicago communities significantly differ from the global mean correlation across all communities. For areas showing negative correlations (Jefferson Park, West Elsdon, and Oakland), we tested if their strong inverse relationships between programs and crime were statistically significant, while for positively correlated communities , we evaluated whether their positive associations genuinely differed from the city-wide average. In both cases, the p-value indicates the probability of observing correlations of such magnitude if the true correlation were actually equal to the global mean. When the p-value falls below 0.05, we can reject the null hypothesis, suggesting that these communities exhibit genuinely distinct relationships between programs and crime rates rather than variations that could have occurred by chance within Chicago's overall pattern.

```
Global Mean Correlation: -0.088

Hypothesis Test Results:
```

	District	Correlation	p-value	Significant?
0	JEFFERSON PARK	-0.751	0.077	0
1	WEST ELSDON	-0.684	0.095	0
2	OAKLAND	-0.656	0.164	0
3	LOOP	0.638	0.012	1
4	WASHINGTON HEIGHTS	0.594	0.021	1
5	AVALON PARK	0.585	0.024	1

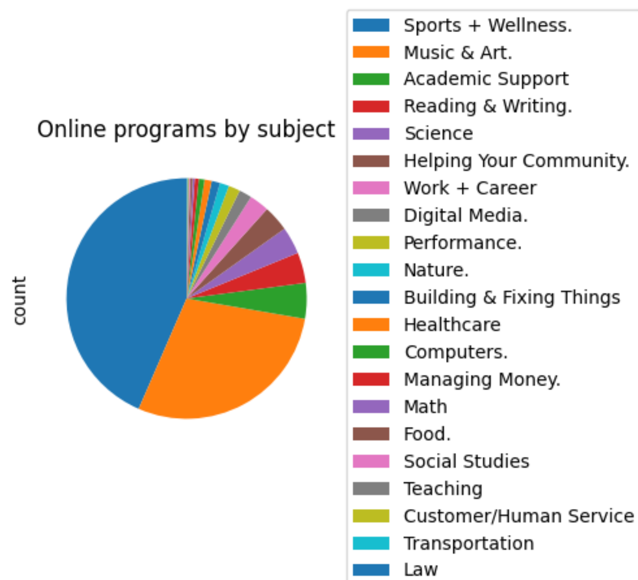


Our key findings are that there seems to be indeed a geniune correlation between crime and availability of programs in some districts which are Loop, Washington Heights, Avalon Park for positive correlation and Jefferson Park, West Elsdon, Oakland for negative correlation. In the case of the positively correlated relationships, the top three communities with the largest correlated all showed statistically significant positive correlations ($p < 0.05$) between crime and program availability. Among them, the loop has the strongest positive correlation, but we cannot rule out that this could very well just be an effect of population and seasonality of crime and programs as this is a dense urban area, and given the vast amount of resources and infrastructure to maintain programs regardless of crime. In the case of the negative correlations, while these show strong negative correlations and are close to being statistically significant, with our p-value we cannot conclusively say that this pattern isn't due to chance.

4.3 Analysis 3: Are there uneven distributions of subject matter programs?

By Daniel Qian

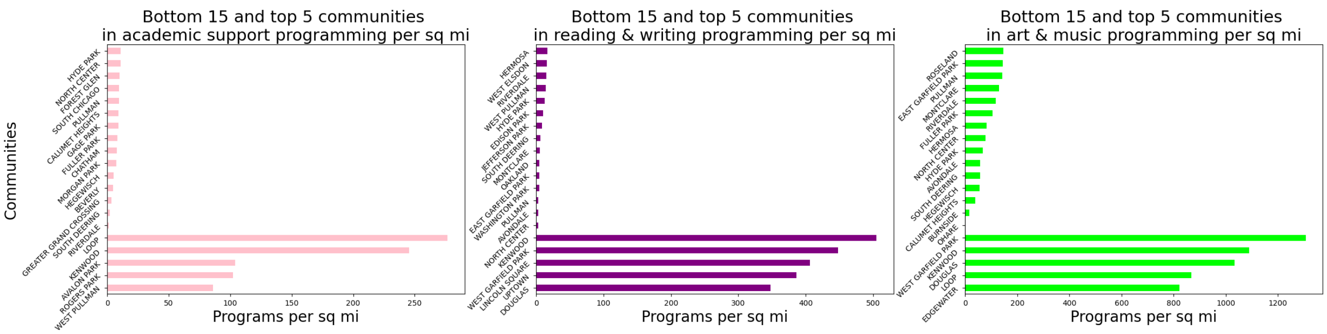
I attempted to find any inequities between program subject matters. I split this task into looking at both online and in-person programs separately. Firstly, I differentiated the online programs by their length using the label 'tag' in the MyChi dataset. I looked at the subject distributions for the one-time events as well as the longer-term programs.



As we can see in the first figure, 'sports & wellness' and 'music & art' take up a large portion of the online longer-term programs. More programming in academic areas such as 'academic support', 'reading & writing', 'science', 'math', etc. could be beneficial because it would provide more lasting support to students who need tutoring.

For the one-time online events, we can see that 'reading & writing' as well as 'music & art' take up a large chunk of the offerings. Adding more diverse programming could benefit communities by providing more opportunities for people to discover new hobbies or interests. For example, offering more computer or digital media events could make people realize they are very interested in coding or filming, photography, etc. and give them a new passion to dive into.

For the in-person programming, I looked for unequal distribution of subjects based on location. To do this, I used a dataset of community boundaries from the Chicago Data Portal in order to assign a community area to each in-person program. This dataset had a "geometry" column that had boundaries for each commmunity in the form of multipolygon complex shapes. First, I used the latitude and longitude columns of the data to turn each program into a point. I then used the geopandas function sjoin to see which community each program fell into. The major issue I anticipated was getting the geographical data of the Chicago Data Portal dataset into a form that was workable with the MyChi data. A problem I encountered was making sure the two datasets were properly aligned so that they could be merged. Eventually I found that the issue was they needed to have the same coordinate reference system. Once that was resolved, I merged the datasets so I could have both the community area as well as the subject of each program. Then, by calculating the square mileage of each community, I was able to find the programs per square mile of each community for different subject matters. Visualizations for a few of the most common subject matters are provided below.

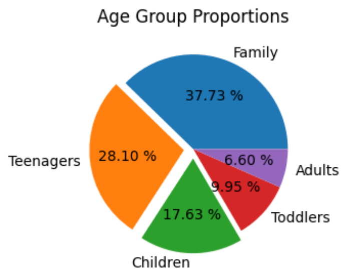


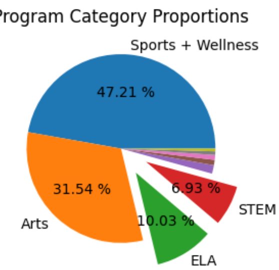
As we can see, many communities are severely underrepresented in the subjects shown and it is reasonable to assume that this pattern of large disparities appears for other subjects as well. Additionally, online programming has a disproportionately large amount of offerings for certain subjects and not enough for many others. It should be noted that this analysis did not include in-person programs with missing latitude and longitude data, which could impact the distributions.

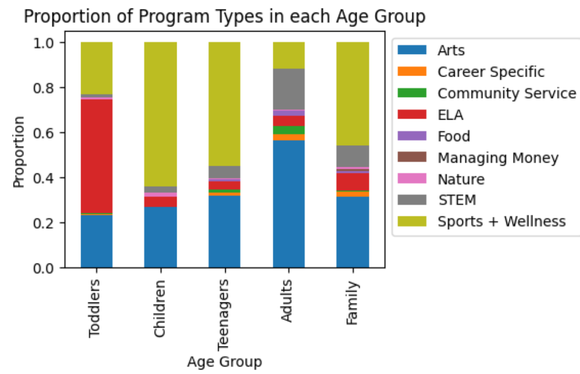
4.4 Analysis 4: How do academically-focused programs affect academic performance?

By Hayden Sterling

The first step in examining the relationship between academic performance and academically-focused programs in Chicago was cleaning and preparing the My CHI My Future Programs Dataset. Programs were categorized into one of 5 age groups. We discarded any programs with a minimum age over 25, and deemed any programs with a maximum age over 25 to be family programs (at the recommendation of the My CHI organization). Programs were also categorized into one of 10 category types, the most important of which for this ELA are STEM (Science, Technology, Engineering, and Mathematics) programs and ELA (English Language Arts) programs. The distributions of these two categorizations can be seen below, along with a barplot further visualizing the distributions.





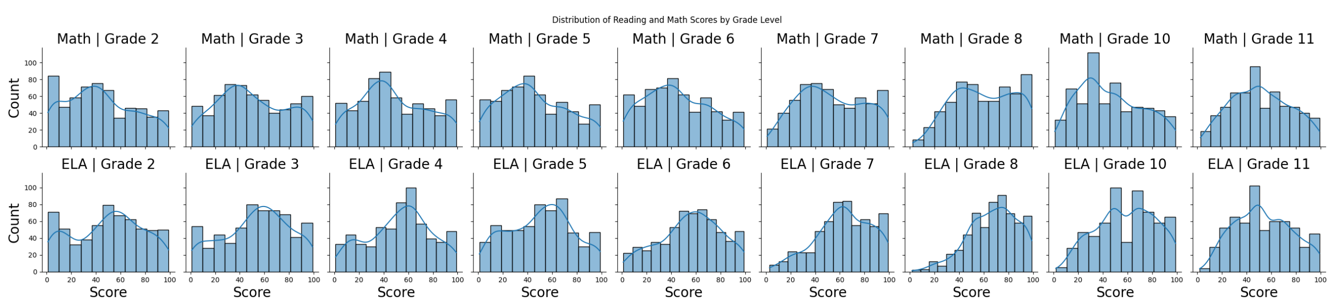


The original ‘Math’ category (which was considered to be STEM) comprised only around 0.03% of programs in the dataset. This is an astoundingly small proportion, considering that Math is a subject that the youth should be focused on in their academic studies.

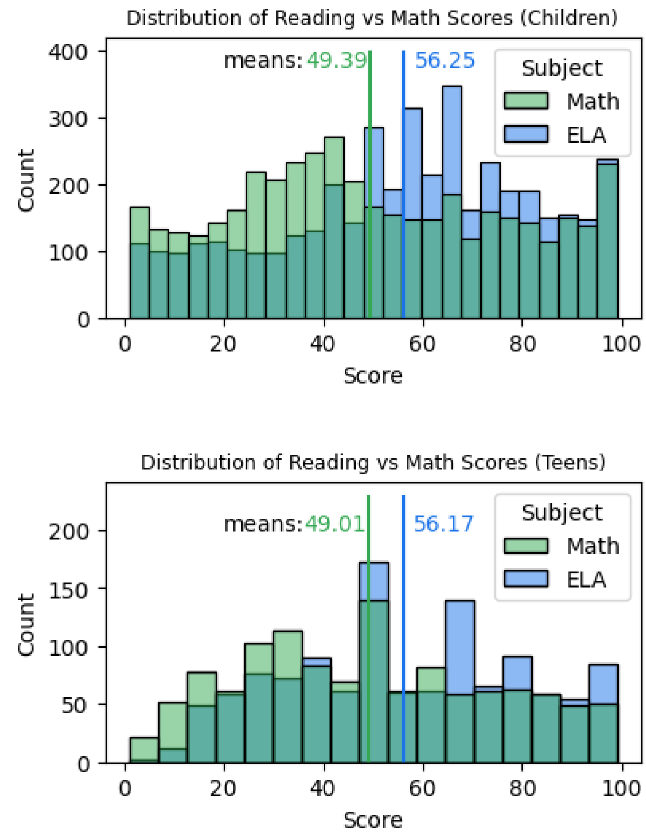
Within the ‘Children’ and ‘Teenagers’ bars it is visible that STEM and ELA programs comprise a fairly small proportion of the overall number of programs meant for children and teenagers. This is negatively affecting equity levels for children and teenagers, as they are most in need of STEM programs because they are in school, and are frequently tested in these subjects.

Next, we wanted to investigate whether or not the frequency of STEM and ELA programs has an impact on academic performance in those subjects for children and adolescents. For this investigation, we will take the number of STEM programs to be a potential indicator of math scores and the number of ELA programs to be a potential indicator of ELA test scores. Although STEM programs may contain subject matter that is not directly related to the material on math exams, research shows that people who are more proficient in STEM possess skills such as quantitative reasoning, logic, and mathematic problem-solving [1], all of which are applicable on a math exam.

The distributions of scores by grade-level and subject from the Progress Reports Dataset are below.

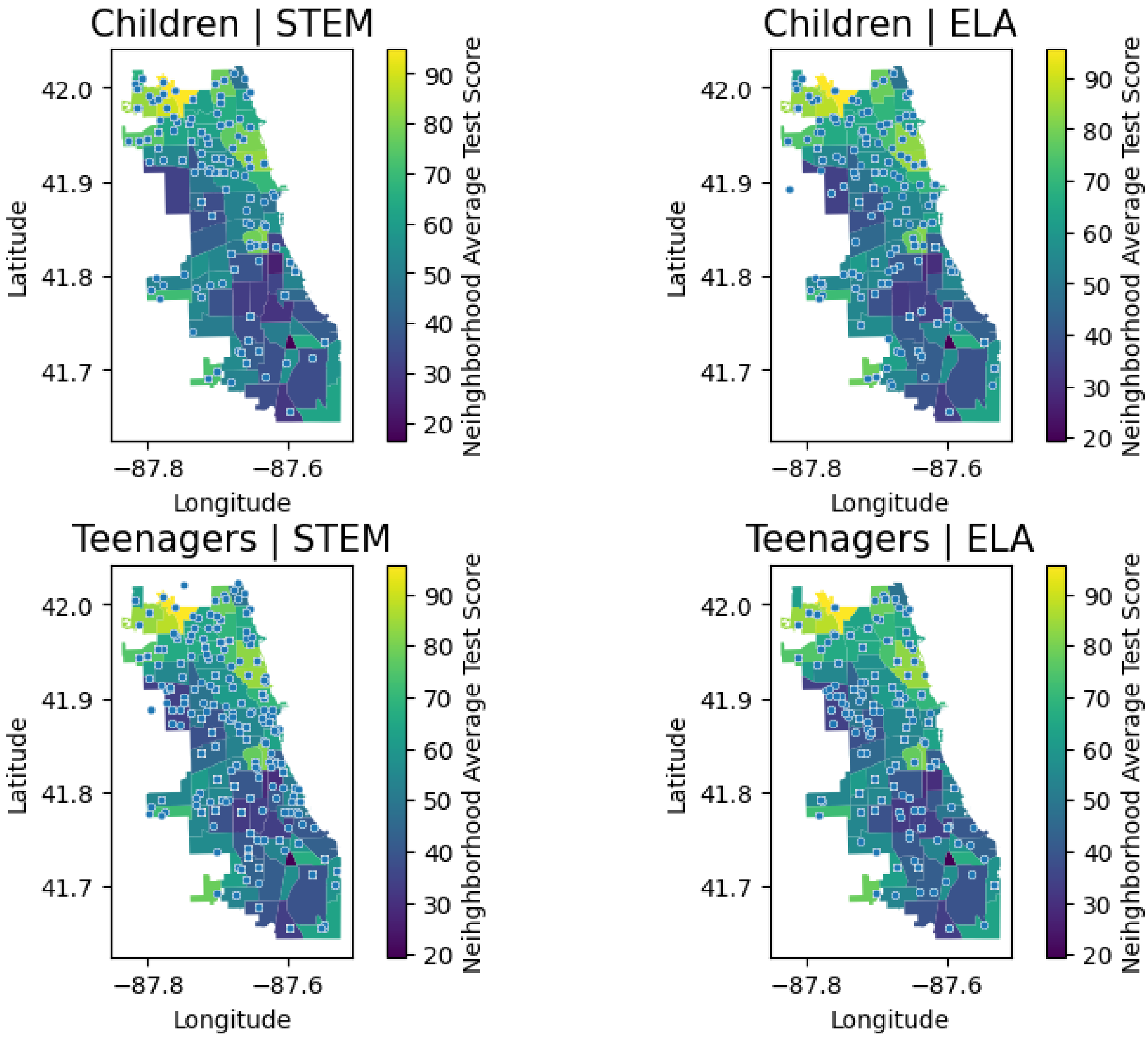


An important trend in these distributions is that math scores tend to be lower than ELA scores for almost all of the grade levels. The lack of math and STEM programs for children and teenagers could be contributing to this. To further investigate, we consolidated the above distributions into two different histograms representing the two age groups. I placed bar plots representing the number of ELA and STEM programs within each respective age group beside these two histograms (viewable on the left side of the plot below).



It is evident that for children, there is a higher number of ELA programs than STEM programs, as well as a higher mean ELA test score. This indicates that there could be an association between the number of academic programs in a subject and academic performance in that subject within the children's age group. When looking at the bottom two graphs representing teenagers, however, it is evident that although a higher mean ELA test score remains true, there is now a higher number of STEM programs. This indicates that the potential association between the number of academic programs in a subject and academic performance in that subject is no longer present in the teenagers' age group.

To investigate that claim even further, we factored in geographic location to the analysis. Below are the locations of programs overlayed on the Chicago map, with color representing mean test score. The plot is again divided by age group and subject level.



For children, in both math scores and ELA scores we see that the Northern side of the city tends to have higher scores, and both STEM and ELA programs tend to be concentrated in the Northern side of the city. For teenagers, however, there is less of an association, as both STEM and ELA programs are fairly evenly distributed around the city while Northern neighborhoods still tend to have higher test scores. To put it in simpler terms, both STEM- and ELA-focused academic programs are fairly effective at raising test scores for children, but not as effective for teenagers.

The challenges that were encountered in this EDA mostly pertained to the data cleaning/preparation phase. The My CHI My Future dataset needed lots of data cleaning. There was some difficulty in determining what qualified as a STEM or ELA program, but this was resolved by sorting programs based on keywords in their descriptions and titles. The Chicago Public Schools Progress Report dataset also needed lots of work before it was clean enough to be operated on. Specifically, imputing was a challenge since there were so many missing values of school's test scores, which had to be imputed using the mean of the same school's test scores.

5 Conclusions

The larger question that this report seeks to answer is how to best address equity issues within the My CHI My Future programs. Generally, factors that indicate the accessibility of programs including their price, transportation, and food, can tend to cause inequity in programs. Crime has been seen to decrease the number of programs in certain neighborhoods of Chicago, also negatively affecting equity levels. Additionally, there is a very high disparity in program density for popular programs, and certain academically-focused programs are found in lower numbers than others across age groups. All of these factors have created large gaps in equity in the My CHI programs. In the next section, we discuss various ways to fix this problem, and thus increase equity levels.

6 Recommendations to stakeholder

The stakeholder for this analysis is the “My CHI. My Future.” Organization. My CHI My Future has recently been concerned with how they can increase equity levels in their youth programs. Through data-backed analysis, we make the following recommendations:

My CHI My Future should eliminate or decrease costs for programs in low income areas. The neighborhood with the highest proportion of paid programs and several more of the top 10 are lower income areas where these fees might prohibit certain people from participating. Additionally, they should increase transportation offerings across all neighborhoods, especially ones farther from downtown Chicago as public transportation is less available and reliable. There is a surprising lack of transportation options for these organizations. Given the wide variety of program types, one might expect more to offer transportation amenities. Furthermore, My CHI My Future should continue their efforts to provide food to organizations in lower-income areas.

Based on our analysis of crime and program relationships across Chicago communities, for densely populated areas like the Loop, we recommend implementing enhanced crime prevention measures and strategically redistributing some programs to neighboring communities, particularly during months with higher crime rates. For residential communities, we suggest developing program sharing networks between neighboring areas, with a focus on increasing availability during periods when nearby high-crime areas experience peaks in criminal activity. To optimize resources, communities should coordinate program schedules, share staff during different peak periods, and establish inter-community communication networks for sharing information about program success and crime patterns.

In order to address unequal distributions for in-person programming, the stakeholder should use the code methods provided to examine which communities are most underserved for each subject. Then, they can reallocate resources to provide more equitable program offerings among Chicago communities. For online programming, the stakeholder could send out surveys to see which subjects are most desired and then tailor their programming to constituents’ demand.

Lastly, My CHI needs to increase the number of math-focused academic programs in Chicago. Given it is an area of common academic struggle, and there is a disproportionately small amount of math programs (0.03% of all youth programs in Chicago), this should be significantly increased. For children specifically, there need to be more academic programs evenly distributed around the city, specifically math programs. For teenagers, the quality of the programs needs to be increased, which would result in better test scores, and thus more opportunity for young adults.

All these recommendations need to be taken into consideration, however they do have limitations. One limitation of this analysis is the limited data that we had access to. Some of our data was old and had large missingness issues. If we received better and more recent data, our analysis could become more accurate.

References

List and number all bibliographical references. When referenced in the text, enclose the citation number in square brackets, for example [1].

[1] Şahin, Ayar, and Adıgüzel. "STEM Education and Its Impact on Students' Problem-Solving and Logical Reasoning Skills." ERIC, The Turkish Online Journal of Educational Technology, October 2021, https://files.eric.ed.gov/fulltext/EJ1312255.pdf.

Appendix

6.1 Chicago Boundaries Dataset

Numeric Variables Distribution

	shape_area	shape_len
count	7.600000e+01	76.000000
mean	7.982215e+07	42697.233573
std	4.401310e+07	13542.979419
min	1.691396e+07	18137.944253
25%	4.953379e+07	31931.454387
50%	7.815582e+07	43125.031081
75%	9.853511e+07	49279.673254
max	3.037971e+08	80389.871800

Variables Count, Unique Values, and Data Type

	Count (Non-Null)	Number of Unique Values	Column Data Type
Column Name			
community	76	76	object
shape_area	76	76	float64
shape_len	76	76	float64
geometry	76	76	geometry

6.2 Chicago Socio-Economic Indicators by Neighborhood from 2010 Census

Numeric Variables Distributions

	PER CAPITA INCOME	Community Area Number
count	78.000000	77.000000
mean	25597.000000	39.000000
std	15196.405541	22.371857
min	8201.000000	1.000000
25%	15804.750000	20.000000
50%	21668.500000	39.000000
75%	28715.750000	58.000000
max	88669.000000	77.000000

Variables Count, Unique Values, and Data Type

Column Name	Count (Non-Null)	Number of Unique Values	Column Data Type
PER CAPITA INCOME	78	78	int64
Community Area Number	77	77	float64
COMMUNITY AREA NAME	78	78	object

6.3 Chicago Crime 2022 Dataset

Numeric Variables Distributions

	District	Community Area	Latitude	Longitude
count	239619.000000	239619.000000	234885.000000	234885.000000
mean	11.310848	36.269273	41.845613	-87.668600
std	7.075555	21.554521	0.088833	0.061010
min	1.000000	1.000000	36.619446	-91.686566
25%	5.000000	22.000000	41.769168	-87.710151
50%	10.000000	32.000000	41.863073	-87.661467
75%	17.000000	53.000000	41.909023	-87.626402
max	31.000000	77.000000	42.022548	-87.524532

Variables Count, Unique Values, and Data Type

Column Name	Count (Non-Null)	Number of Unique Values	Column Data Type
Date	239619	112310	object
Primary Type	239619	31	object

Column Name	Count (Non-Null)	Number of Unique Values	Column Data Type
Description	239619	286	object
District	239619	23	int64
Community Area	239619	77	int64
Latitude	234885	118321	float64
Longitude	234885	118305	float64

6.4 Chicago Public Schools 2018-19 Progress Report Data

Numeric Variables Distribution

The variable names correspond to {Subject}_{Grade level}, so for example, ELA_2 contains the English Language Arts scores for 2nd graders across schools. Here, we print a small number of the variables since there are too many to fit on the page.

	ELA_2	Math_2	ELA_10	Math_10
count	451.000000	452.000000	132.000000	132.000000
mean	47.046563	44.090708	72.045455	30.537879
std	31.553694	30.958641	17.596349	19.961186
min	1.000000	1.000000	18.000000	1.000000
25%	18.000000	15.000000	66.000000	16.000000
50%	47.000000	42.000000	72.500000	29.000000
75%	76.000000	72.000000	83.000000	34.000000
max	99.000000	99.000000	99.000000	95.000000

Variables Count, Unique Values, and Data Type

Column Name	Count (Non-Null)	Number of Unique Values	Column Data Type
ELA_2	451	97	float64
Math_2	452	97	float64
ELA_10	132	29	float64
Math_10	132	28	float64

This data has missingness issues across all of the variables. The dataset contains information on 654 schools, and some of the variables only have 135 non-missing values. However, we will not need to impute some of these. For example, the last four variables in the second table are for high school grades. The reason that there are only ~130 observations could be because there are only ~130 high schools in the dataset, so not all rows will have information for those variables if they are not high schools.

EDA 3: Crime and Activity Outliers

```
Crime Outliers (Communities with unusually high or low crime counts):

Activity Outliers (Communities with unusually high or low activity counts):

Summary Statistics:
```

	community	crime_count
5	AUSTIN	11974
47	NEAR NORTH SIDE	10335
49	NEAR WEST SIDE	9304
64	SOUTH SHORE	8213

	community	activity_count
34	IRVING PARK	5482
57	PULLMAN	3417
49	NEAR WEST SIDE	2206
45	MORGAN PARK	1974

	Metric	Value
0	Total communities with outlier crime counts	4
1	Average crime count across all communities	3051.4
2	Median crime count across all communities	2179.0
3	Total communities with outlier activity counts	4
4	Average activity count across all communities	812.5
5	Median activity count across all communities	700.0